Project Brief: Rombus Software Development

Project Overview:
Rombus currently utilizes an in-house Python-based plugin for Blender that facilitates the creation of 3D print-ready files from uploaded human anatomy scans. This plugin leverages core Blender operations and geometry nodes to generate parametric 3D models, which can be updated in real-time based on device specification changes.

Project Goal:
The primary objective is to transition the existing Blender plugin into a cloud-based, customer-facing software platform. The new platform should be modular, allowing for the easy addition or removal of features (e.g., design templates, anatomical corrections) to meet client-specific needs. Additionally, the platform should incorporate machine learning capabilities and enable integration with various app-based features, such as 3D scanning and automatic anatomical landmark recognition.

Devices to be Integrated:
- Foot insoles
- Hand splints
- Leg splints
- Spinal braces

Key Requirements:

- Cloud-Based Platform:
    - Transition the current Blender plugin into a robust cloud-based solution.
    - Ensure modularity for easy feature customization, including design templates and anatomical corrections.

- Machine Learning Integration:
    - Implement machine learning algorithms to enhance the accuracy of 3D designs and anatomical corrections.

- App Integration :
    - 3D Scanning: Integrate with 3D scanning devices for seamless uploads.
    - Automatic Landmark Recognition: Develop functionality to automatically recognize and mark key anatomical landmarks on uploaded scans.

Software Workflow:

1. Users log into the platform.
2. Users scan the anatomy using integrated in-app 3D scanning and upload it to the platform.
3. Users fill out an order form with device specifications; the CAD model updates in real-time based on the form inputs.
4. Provide the option for manual design adjustments if necessary.
5. Submit the order for 3D printing by Rombus or offer the option to download the file after payment.
6. Implement a pay-per-use payment structure.
7. Include order tracking functionality.

Examples of Similar Software:
Example 1
Example 2
Example 3
Example 4
Example 5
Example 6
Example 7

Timeline:
We anticipate the project to take approximately 4 to 7 months from commencement to completion.

Budget:
As we are in the early stages of development, we request your proposed approach to software development, including a breakdown of costs. While we are seeking a cost-effective solution, we are open to exploring various options based on your recommendations for best practices and scalability.